Evaluation Report: Two-Stage Q&A Generator Output

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Target Document: EasyMetagenome: *A user-friendly and flexible pipeline for shotgun metagenomic analysis in microbiome research* (Bai et al., 2025)

Method Tested: Two-Stage Generation (5 Broad Questions - 5 Technical Follow-ups)

Overall Conclusion: HIGHLY EFFECTIVE. The Two-Stage method successfully translates general concepts into deep technical facts.

A. Technical report: Script flaws – Debugging and Validation:

This part will outline the transformation of the initial script following the encountered errors.

The process encountered 2 layers of critical errors which were addressed through code modifications.

Layer 1: API protocol mismatch

"Error: 'choices'"

The script used the OpenAI chat format "messages" as input and expecting "choices" as the output structure, but the GPT-OSS server did not use the chat format, it required a simpler completion format "single 'prompt' string".

I modified the call_model function and rewrote it to combine the system and user instructions into a single 'prompt' matching the API requirement. Replaced "messages": [...] array with the "prompt": full_prompt key.

Layer 2: Streaming Failure

"Error: Extra data: line 2 column 1 (char 114)"

The server outputs chunks of JSON using Server-Sent Events, which the r.json() in the script cannot parse.

I changed the "requests.post" to "stream=True" and used "r.iter_lines()" function to process the stream line by line and concatenate the "response" chunks.

B. <u>Context and Factual accuracy:</u>

<u>Part 1: Effectiveness in extracting technical depth:</u>

The goal of this method was to move past summaries and extract actionable, technical facts such as tools, versions, parameters … and the script succeeded completely here.

Example:

Broad Q: What are the main features and functionalities of EasyMetagenome that differentiate it from existing metagenomic pipelines?

Answer: A detailed listing of features like **integrated host-read removal**, **dual analytical tracks**, and **Snakemake reproducibility**.

Excellent **Conceptual Alignment** data that teaches the LLM the high-level justification for the agent.

Follow-up Q: *Could you detail the mechanism by which the pipeline tracks and locks the versions of all key components—such as the Kraken2/Bracken taxonomic database, the HUMAnN3 functional database, the MEGAHIT/metaSPAdes binaries, and the Snakemake environment itself?*

Answer: A highly detailed breakdown of the **Git/Conda/Docker provenance system**. Explicitly names files like config.yaml, environment.lock.yml, and db_metadata.yaml used for versioning.

Perfect **Tool and Configuration Grounding** data. This is the ideal input for teaching the agent reliability and traceability.

Follow-up Q: *How does EasyMetagenome integrate results from multiple binning algorithms... and what specific parameters or filtering criteria does it use to optimize completeness versus contamination in the final bins?*

Answer: Detailed protocol for using **DAS-Tool** to merge **MetaBAT2/MaxBin2/CONCOCT** outputs, followed by **CheckM** and **dRep** filtering using the industry-standard **90% completeness / 5% contamination** thresholds.

Excellent **Workflow Execution** data. Provides the numerical parameters required for the final decision-making logic of our agent.

<u>Part 2: Factual accuracy and Hallucination:</u>

Demonstrated strong factual grounding, the use of the 8,000-character context window is sufficient for deep extraction.

Answers were free of hallucinations regarding tool names, associated software and analytical tracks.

Minor weaknesses: in one of the follow-up questions the model proposes solutions that were listed as optional post processing in the paper rather than the default workflow, it shows that the LLM is good at suggesting best practice but still requires tighter prompt control to differentiate between a default method and a more advanced solution.

<u>Part 3: Suggested refinements for better SFT suitability:</u>

The tools and parameters were in paragraph format rather than a list. Especially for the parameters list, it would be better if we enforce a dedicated JSON field for key parameters to train the model's conditional reasoning logic. (e.g., *If low-coverage, use this parameter*) [cite: The precise extraction of the numerical values (90%, 5%, etc.)

Lastly, the metadata does not include a source URL/DOI, if we are planning to link the Q&A pairs back to the origin for RAG attribution, we need to integrate a field to capture the paper's main DOI or URL.